
Ravindra Sanjay Lahire



At Post.Bhalur Tq.Nanadagon Dist.Nashik Maharashtra 

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Objective

Experienced Embedded Firmware Engineer with 3 years of expertise in developing and optimizing firmware for various organizations. Adept at designing, implementing, and testing embedded systems to enhance performance and reliability. Seeking to leverage my skills and knowledge to contribute to innovative projects and drive technological advancements To work in environment where I can use my knowledge and my skills to Empower the whole organisation and also to develop myself in multidisciplinary in the industry.

Experience

- Rishabh Instruments group Mumbai** August 2023 - Still working
Embedded Firmware Developer
Working as New Product Development Roll and Responsibilities in Firmware software developer Device Driver Development Life cycle Development in Embedded product

1) Project : Oil and Gas Dispenser (Microcontroller RA6M5)

Roles and Responsibilities: Embedded Development in C, C++, USB Device, data structures and multi threading.with Azure RTOS handling Thrdx, scheduling and CAN protocol (different message id) , Mutex and Semaphore

- UI design and development CSD data Base
- Handling Thrdx and Error (throughout system in code unpredictable error occurred handling)
- Printers communication design and development with Main CPU side with integrated in product embedded protocols (I2C,SPI,UART, CAN),Signal processing , ADC,DAC and interrupt handling

Designing & Optimization of Firmware/Software architecture for different embedded Products

Tools : QT,E2studio, STM32cubeMX, STM32CubeIDE, Docklight Beyond compare

2) Absolute and Differential Pressure meter

Role and Responsibilities:- Device Driver progmming Designed (Structures and Functions Based) Communication with Microcontroller and GZP190 reading Analog and conversion Decimals Format

programming (SD93F115) UART, I2C

Microcontroller :- SD9F115 , UART Communication, I2C Communication GZP190 Air Pressure sensor

Tools:- CDK, STM32CUBIDE, STM32CubeMX, Keil uVision5

Extra: Design and Development custmation microcontroller and SOC

3) Single Phase Preventer:

Design and development life cycle and Development Device Driver progmming and various products testing process

STM32F4 series controller and UART communication

Tools:- STM32CUBEMX STM32CUBIDE

4) Earth Tester:

4G Device Driver Development (structure and function based)

Responsibility - as EC200 quectel(4G) Module Communication with STM32 Controller and continue monitring Soil Resistivity

familiar and hand's on experience process and thread with STM32 Controller mange ISR UART Protocol, I2C, SPI,CAN, OCP, TCPIP, USB Driver websocket, various peripheral experience

ARM Cortex series controller: familiar with

STM32, AVR, PIC NXP, Arduino,LPC2148 , HAL library, MCAL library Device Drivers programming, Linux kernel OS programming, c++ language programming, JTAG data structure and algorithm, object oriented programming language,

- **People Tech Group in Hyderabad**
System Engineer (Embedded developer)

September 2022 year - August 2023

Project:

1)Smart car door lock System

Role & Responsebilty : Communication CAN to CAN Communication with each ECU and Send signals Dashboard using STM32 control Programming. Device driver programming and Embbbbed c programming

2) Tire Temperature Measurement

Role & Responsebilty : UART Communication protocol Device Drivers using STM32 controller programming and CAN Protocol follow standard (ISO 11898) and c programming language

3) Advanced Driver Assignment System

Role & Responsebilty : Flexray and CAN Data PDU Routers and RTE layers, Communication ECU as per Priority Bases and Interrupt handler programming. Follow AUTOSAR Standard c programming and c++ programming language

4) Automation Testing : IPC Instrument Cluster control

Role & Responsebilty : working on Replacement Jenkins and Generate New Jenkins Test on cluster or Binery Replacement Infotainment Basic functionality checking. Flashing new build different VCU using QNX cross check with CANOE tools

Create test plans

Sanity ech and every functionality like map, mode, fuel level, speedo metre had all functionality

Skill :

ARM Cortex series controller: familiar with

STM32, AVR, PIC NXP, Arduino,LPC2148 , HAL library, MCAL library Device Drivers programming, Linux kernel OS programming, c++ language programming, JTAG data structure and algorithm, object oriented programming language,

Protocol: CAN, UART, SPI, UDS, CAN shlid, Ethernet, I2C , TCIP, LIN, CANOE

Languages:

C language

Embbbed C

C++ language

AUTOSAR

Data structures and Algorithms

Object Oriented Programming

Tools: IDE GM simulator

STM32 cub IDE

MP lab ide

Arduino ide

Keill 5

CANOE

CAN Ays

QNX , QFIL , Fash helpers,GM simulator

Education

- **RSBSS School of Aurangabad**
10(SSC)
54.80

2012

- **RSBSS School of Aurangabad** 2014
12 (HSC)
58.31
- **MIT Polytechnic college Yola** 2016
Diploma in Electronics and tell- Communication
70.29
- **SND COLLEGE OF Engineering and Research Centre Yola)** 2020
BE(Electronics and tell-Communication)
6.16

Skills

Embedded C C++ Linux device driver programming Data structures
 CAN UART I2C UDS Communication protocol AUTOSAR STM32 Controller
 Object oriented and programming language
 EMBEDDED SYSTEMS linux commands, ARM-gcc Data Structures c, c++,Tools-keil ,IAR,
 Pheripheral Driver timer,WDT RTOS AVR,UART,SPI I2C,CAN,communication protocol
 Microcontroller 16,32 bit

CERTIFICATIONS

- **Advanced Embbbbed System Course**
Vector India in Hyderabad
Duration=06 Month.

Projects

- **Accelerometers Based using Digital pen**
Project in Diploma
- **Smart Helmate Alcoholic Dector and Accident Alert Using GSM and GPS Bike Control Ridder**
Project in BE final year
- **Touch Screen Based Device Control system**
Project in vector India Teraning institute in Hyderabad

Achievements & Awards

- Indian Army SSB Interview (7 Days) Final Conference Round out in Allahabad 2 Times
- Arms Forces Preparatory Academy (Nagpur) 1) Screening Test 2) Psychology Test 3) Group Testing

Publications

- **International Research Journal of Engineering and Technology (IRJET)**
Samart Helmet Alcohol Detector Accident Alert using GSM and GPS Bike Control Rider

Interests

- Foreign policy
- Psychology
- Arts and culture
- International Politics

Activities

- Participation in sate level Poster Presentation Competition at matoshri Institute of Technology, yeola
- Participation in National Robotics champ.organised by Gadets and matoshri Institute of Technology,Yeola

- Participation in Technical Quiz Competition at SND polytechnic, Yeola

LANGUAGES

- English
Hindi
Marathi

PERSONAL PROFILE

- Name : LAHIRE RAVINDRA SANJAY
Date of Birth: 09/06/1996
Mo.No : 7499294217,9890739086
Blood Group: O negative
Marital status : Unmarried
Nationality : Indian Know
language : English, Hindi, Marathi
Permanent Address : At Post.Bhalur Tq.nanadgon Dist.Nashik Maharashtra 423104